

Ultrafiltration

Pre-Lab Plan
CHE 382 Lab 4
Due: March 14th, 2019



Group # 1
Thursday, PM section

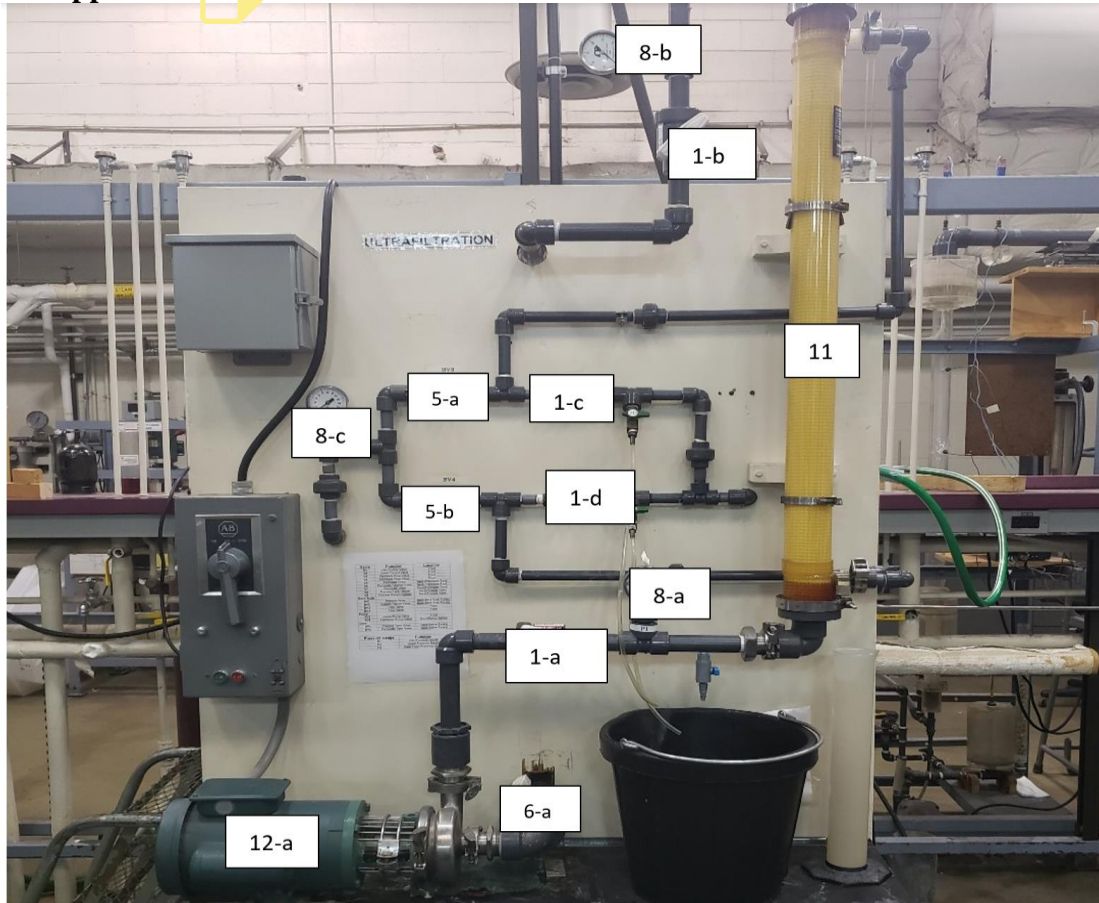
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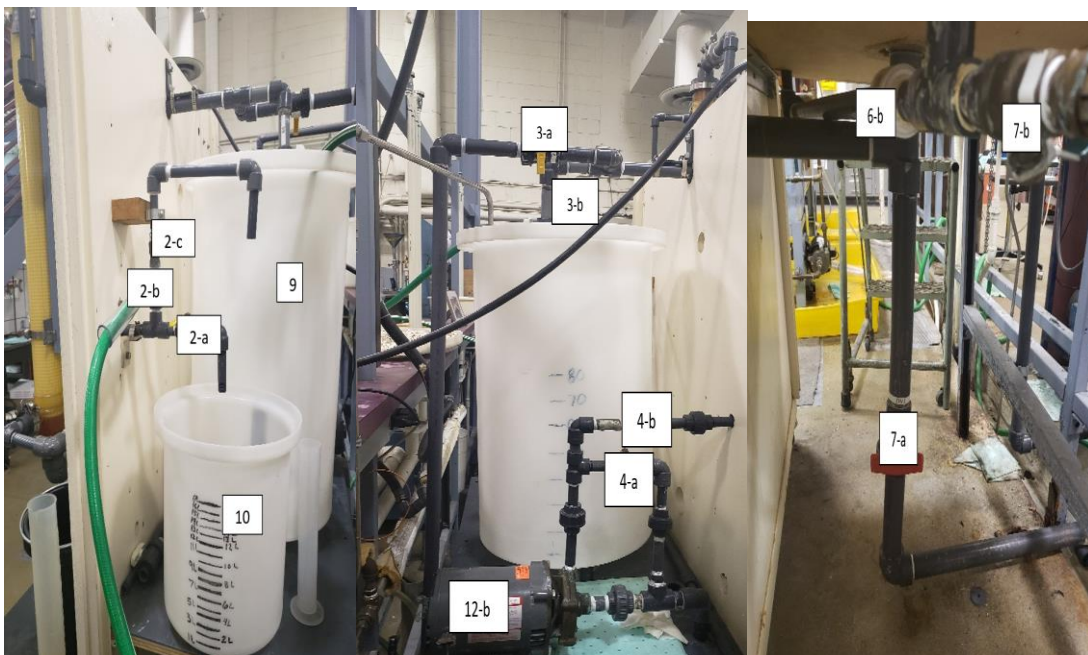
1. Experimental Objective

The purpose of this laboratory experiment is to examine an ultrafiltration system and its filtering effectiveness. The experiment based on measurements of the flux through the membrane for different pressure drops and for different milk concentrations. Ultrafiltration is a fairly new process (1970's) and is useful to concentrate solutions containing small macromolecules.

2. Experimental

2.1 Apparatus





#	Name	Description
1(a-d)	Gate valve (V1-V4)	Gate valve used for flow control of different piping located on the front of the apparatus
2(a-c)	Gate valve (V5-V7)	Gate valve used for flow control of different piping associated with the permeate tank located on the back of the apparatus
3(a-b)	Gate valve (V8-V9)	Gate valve used for flow control of different piping associated with the process tank located on the back of the apparatus
4(a-b)	Gate valve (BV1-BV2)	Gate valve used for flow control of piping associated with the back flush pump located on the back of the apparatus
5(a-b)	Gate valve (BV3-BV4)	Gate valve used for flow control of piping located on the front of the apparatus
6(a-b)	Gate valve (PV1-PV2)	Gate valve used for flow control of piping located upstream of pumps located on front and back of the apparatus
7(a-b)	Gate valve (DV1-DV2)	Drain valves used for flow control of the drain from tanks located on the back of the apparatus
8(a-c)	Pressure Gauge(P1-P3)	Pressure measurement instrumentation used for pressure monitoring of inlet, outlet and back flush streams
9	Process Tank	Plastic, graduated, atmospheric storage container used to accumulate volumes of process fluid

10	Permeate Tank	Plastic, graduated, atmospheric storage container used to accumulate volumes of permeate
11	Membrane Filter	Cylindrical membrane filtration unit used in the separation process located on the front of the apparatus
12(a-b)	Pump(2)	Electrically power process equipment used to increase head of process fluid

2.2 Materials and Supplies

Name	Description
Milk	To measure different milk concentrations (Whole, 2%, 1% or Skim milk)
Deionized water	Using 105°F deionized determine the permeate water flux
Graduated cylinder	For purpose of a measurement of flux
Phosphoric acid	To make phosphoric acid solution in deionized water for acid cycle cleaning
Sodium Hydroxide	To make up 20 L of a 1% sodium hydroxide solution in deionized water at 105°F for caustic cycle cleaning
Stirring rod	To stir solutions
Bleach	For sanitation cycle
Litmus paper	To measure pH of prepared solutions for cleaning procedures

2.3 Experimental Plan

I. Cleaning Procedure

(1) LIQUID SOLUTION FLUSH

- A. Fill the process tank with 20 L deionized water at approximately 105°F. Open the Intake-Pump valve (PV1).
- B. Direct the permeate outlet flow to the process tank.
- C. Close the inlet pressure control valve (V1) and open the outlet pressure control valve (V2).
- D. Start the circulation pump.
- E. Slowly open the inlet pressure control valve (V1) until a pressure of 20 psi is obtained on the adjacent pressure gauge (P1).

- F. Slowly close the outlet pressure control valve (V2) until a pressure of 8 to 10 psi is obtained on the adjacent pressure gauge (P2).
- G. Readjust the inlet pressure control valve (V1) until the inlet pressure is stabilized at 25 psi and the outlet pressure is stabilized at 10 psi.
- H. Allow the system to run for 10-15 minutes. During this time check the system for leaks. There should be NO AIR BUBBLES visible in the tank. If air bubbles are present, check with an instructor.
- I. Slowly open the outlet pressure control valve (V2) fully.
- J. Close down the inlet pressure control valve (V1).
- K. Stop the circulation pump.
- L. Drain the water from the process tank and filter.
- M. Rinse the process tank.

(2) ACID CYCLE

- A. Make up 20 L of approximately pH 2.00 phosphoric acid solution in deionized water at 105°F. ($K_{P1} = 7.11 \times 10^{-3}$).
- B. Repeat steps B to G of the Liquid Solution Flush
- C. Run in this process mode for 10 minutes. Next close off the permeate flow valves (V3) (V4) and run the acid solution through the filter for another 5 minutes.
- D. Repeat steps I to M of the Liquid Solution Flush

(3) DRAIN AND RINSE CYCLE

- A. Refill the process tank with 50 liters of 105°F deionized water.
- B. Direct the permeate outlet flow to permeate tank.
- C. Open Permeate Drain Valve (DV2).
- D. Repeat steps C to G of the Liquid Solution Flush. Momentarily crack the drain valves during the rinse cycle to flush out any trapped acid solution. Make sure to stop the pump before the water level in the process tank falls below the level of the inlet pipe to the pump.
- E. Rinse Permeate Tank

(4) CAUSTIC CYCLE (Save Permeate)

- A. Make up 20 L of a 1% sodium hydroxide solution in deionized water at 105°F. Use weight basis (density of water is 8.33 lbs/gallon).
- B. Repeat steps B to M of the Liquid Solution Flush, except direct the permeate to the permeate tank instead of the process tank. Make sure the drain valve on the permeate tank is closed. The permeate solution is to be saved for backflushing.

(5) BACK FLUSH MODE

- A. For the permeate area, close the permeate recycle valve (V6). Open the permeate pump valve (PV2). Close the permeate tank drain valve (DV2). Open the permeate drain valve (V5). In the back flush area, close the recycle back flush valve (BV1) and open the back flush return to system valve (BV2). In the front, open the lower left back flush flow valve (BV4), the upper left backflush flow valve (BV3) and output process valve (V2). All other valves in the front should be fully closed except Intake Pump Valve (PV1).
- B. Turn on back flush pump.
- C. Allow the permeate liquid to flow from permeate tank, through the system via back flush, and into the process tank).
- D. Once the permeate tank level halves, Close the process tank drain valve (DV1).
- E. Drain the permeate tank, close all back flush valves, and open outlet pressure valve.
- F. Turn off back flush pump, turn on circulation pump.
- G. Slowly open the inlet pressure valve to 20 psi.
- H. Slowly close the outlet pressure control valve to 10 psi.
- I. Adjust the valves until the inlet is set to 25 psi and outlet is set to 10 psi

- J. Run the system for 15 minutes
- K. Check for any leaks in the system or air bubbles in the tank
- L. Slowly open the outlet pressure valve and close the inlet pressure valve
- M. Drain the filter and the process tank
- N. Rinse the process tank to prepare for the next cycle

(6) SANITATION CYCLE

A. Make up a 200 ppm (200 ppm available chlorine) sodium hypochlorite solution in 20 L of deionized water at 105°F. (Use 350 mL household bleach/50 L water.)

B. Repeat steps B to M of the Liquid Solution Flush
Repeat the drain and rinse cycle (3) described above.

II. Experimental Procedure

WATER FLUX TEST

After the cleaning procedure has been completed, determine the permeate water flux of the ultrafilter at an inlet pressure of 25 psi and outlet pressures of 5, 10, 20, and 15 psi using 20L of 105°F deionized water.

MILK TEST

- A. Fill the process tank with 80 liters of 105°F deionized water. Add 1 gallon of milk and stir.
- B. Start the system in the normal manner and adjust the pressure to 25 psi inlet and 5 psi outlet.
- C. Run in this manner until 10% recovery is reached.
- D. Repeat the experiments at recoveries of 25% and 40%.

Project Safety Evaluation

Project Title: **Ultrafiltration**

Date: 3/14/2019

	<u>Yes/No</u>	
Potential electrical hazards?	<u>Yes</u>	If yes, identify conditions:pumps, junction box 
Potential mechanical hazards?	<u>Yes</u>	If yes, identify conditions:pumps 
Potential pressure hazards?	<u>Yes</u>	If yes, identify conditions:not higher than 25 psia
Potential temperature hazards?	<u>Yes</u>	If yes, identify conditions: 40.65 C 
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Phosphoric acid, Sodium Hydroxide,Bleach
Gloves required?	Yes	If yes, specify type: Chemical resistant gloves 
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Phosphoric acid, Sodium Hydroxide, Bleach
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	40.55	
Pressure (psia)	25	
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Yes	Pressure gauges, thermocouple on inlet water, valves 

I have read relevant background material for the Unit Operations Laboratory entitled:
“Ultrafiltration” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Ekaterina Ivanov

_____	03/14/2019	_____
signature of team member	date	

David Schulz

_____	03/14/2019	_____
signature of team member	date	

Raghad Albadri

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signature of team member	date	

Irene Jose

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